

Assignment 2: Data Ingestion & Data Warehouse Modelling

Onboarding Flow Optimization A/B Test

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Abstract

This document specifies the design and implementation of the data platform that transforms the raw event stream defined in Assignment 1 into an analysis-ready warehouse and data mart. It follows a four-layer (“medallion”) architecture — Raw, Staging, Warehouse, Data Mart — implemented in SQLite for portability, and documents the rationale behind each design decision: why this layering rather than a single flat table, why a dimensional (star-schema) model rather than a normalized transactional model, what the ingestion and deduplication contract is, and how data quality is verified rather than assumed. Every SQL artifact referenced here is version- controlled under `schema/` and executed, in order, by `build_db.py`.

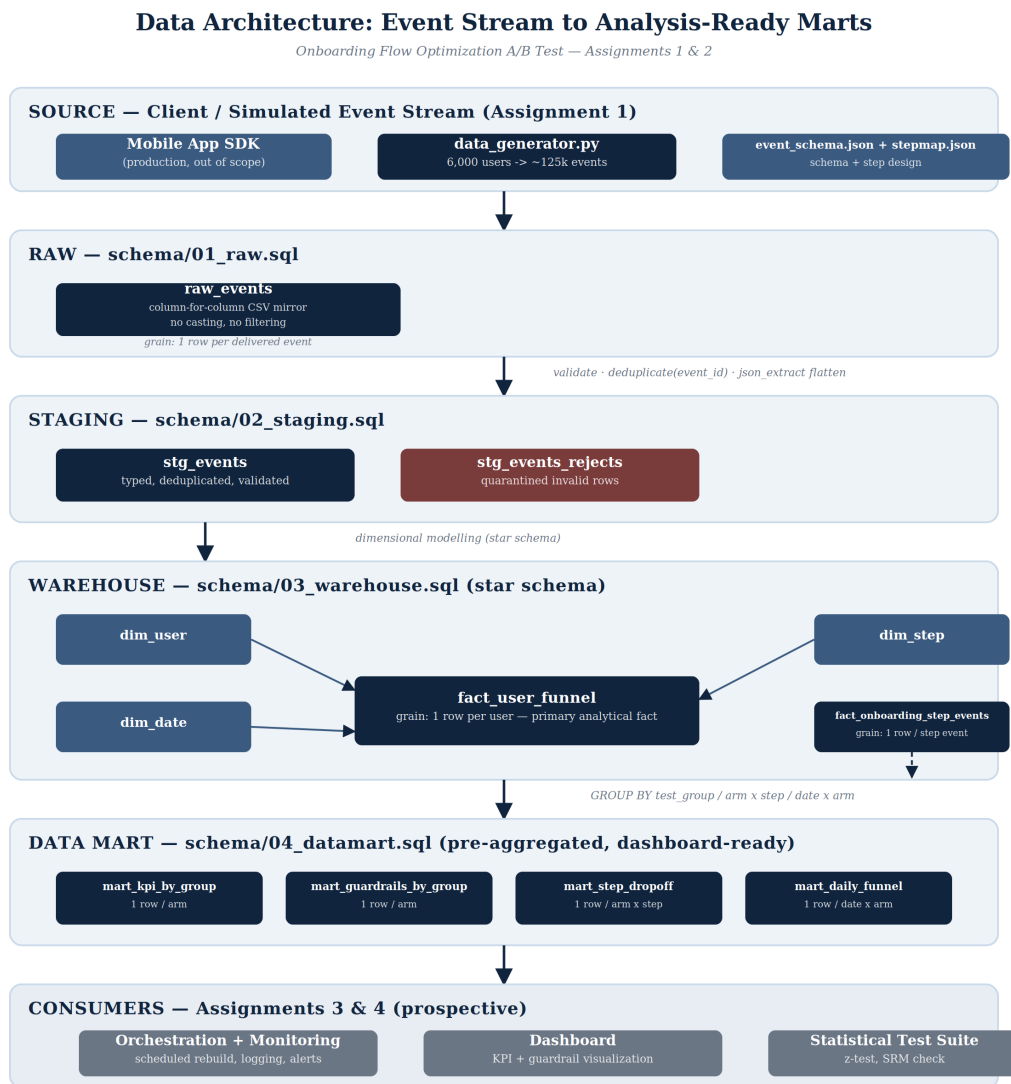
1. Introduction

Assignment 1 produces an append-only, long-format event stream: one row per user action, timestamped, tagged with experiment arm and platform. That format is optimal for *capture* — it can represent any event type without a schema change — but is poor for *analysis*: computing a single KPI such as “onboarding completion rate by arm” against the raw event table requires a self-join and conditional aggregation that is expensive, error-prone to repeat correctly across every metric in the KPI framework, and opaque to a non-technical dashboard consumer. This assignment closes that gap.

2. Architectural Overview

2.1 Layer Summary

```
raw_events                                RAW
| (1) validate (2) deduplicate (3) flatten JSON -> typed columns
v
stg_events, stg_events_rejects            STAGING
| (4) dimensional modelling (star schema)
v
dim_user, dim_date, dim_step,
fact_onboarding_step_events,
fact_user_funnel                          WAREHOUSE
| (5) group-by aggregation to KPI/guardrail grain
v
mart_kpi_by_group, mart_guardrails_by_group,
mart_step_dropoff, mart_daily_funnel      DATA MART
```



Data architecture: event stream to analysis-ready marts

Figure 1. Full pipeline, source through prospective consumers. Every layer boundary corresponds to exactly one `schema/*.sql` file (Section 3).

2.2 Rationale for Layering

A single-layer design — transforming raw JSON events directly into the final KPI tables in one pass — was considered and rejected for three reasons:

- Debuggability.** When a KPI number looks wrong, a layered pipeline lets the analyst inspect intermediate state (did the row survive staging? does it appear in the warehouse fact table? does the mart aggregation match a manual `GROUP BY` against the fact table?) rather than re-deriving the entire transformation from raw JSON to find the defect.
- Reusability.** `fact_user_funnel` (Warehouse layer) is the single source of truth for every KPI and guardrail in the Assignment 1 KPI framework. Without it, each new mart table would need to independently

re-derive funnel state from raw events, multiplying the surface area for transformation bugs and definitional drift between marts.

3. **Auditability of data quality.** Validation and deduplication belong at a single, well-defined boundary (Raw to Staging) rather than being re-implemented, potentially inconsistently, in every downstream query.

2.3 Rationale for a Dimensional (Star Schema) Model

The warehouse layer uses a **star schema** — a central fact table (`fact_user_funnel`) at a fixed grain, joined to descriptive dimension tables (`dim_user` , `dim_date` , `dim_step`) — rather than a fully normalized (3NF) transactional model. This follows standard data- warehousing practice (the Kimball dimensional-modeling approach) for analytical workloads, for two concrete reasons specific to this dataset:

- **Query simplicity for the consuming layer.** Every KPI in the Assignment 1 framework is a `GROUP BY test_group` aggregation over a small number of boolean/flag columns. A star schema with one wide fact table at “one row per user” grain makes this a single-table aggregation; a normalized model would require reconstructing that same row via joins across several transactional tables at query time, for every dashboard query.
- **Fixed, known grain.** Because the business question in Assignment 1 is stated entirely in terms of per-user funnel outcomes (did this user complete onboarding, activate, return, purchase), the natural grain of the primary fact table is unambiguous: one row per user. A normalized model optimizes for transactional write consistency, which is not the workload here — the data is immutable, append-only event history.

The star schema is not applied dogmatically: a second, more granular fact table (`fact_onboarding_step_events` , grain: one row per step-level event) is retained alongside the summary fact, because step-level drop-off analysis (used by `mart_step_dropoff`) genuinely needs event-level, not user-level, granularity. This is a deliberate two-fact-table design rather than forcing a single grain to serve both purposes.

3. Layer-by-Layer Design

3.1 Raw Layer (`schema/01_raw.sql`)

`raw_events` is a column-for-column mirror of the source event CSV, plus two ingestion-metadata columns (`_ingested_at` , `_source_file`). No casting, filtering, or validation happens here — the raw layer’s sole responsibility is to be a faithful, replayable copy of what ingestion received, including any malformed rows, so that a staging-layer bug can always be diagnosed by re-running staging against an unmodified raw capture rather than needing to re-ingest from source.

Indexes on `event_id` and `user_id` are created at this layer purely to keep the staging deduplication query (Section 3.2) performant; they do not imply any constraint enforcement, which is intentionally deferred to staging.

3.2 Staging Layer (`schema/02_staging.sql`)

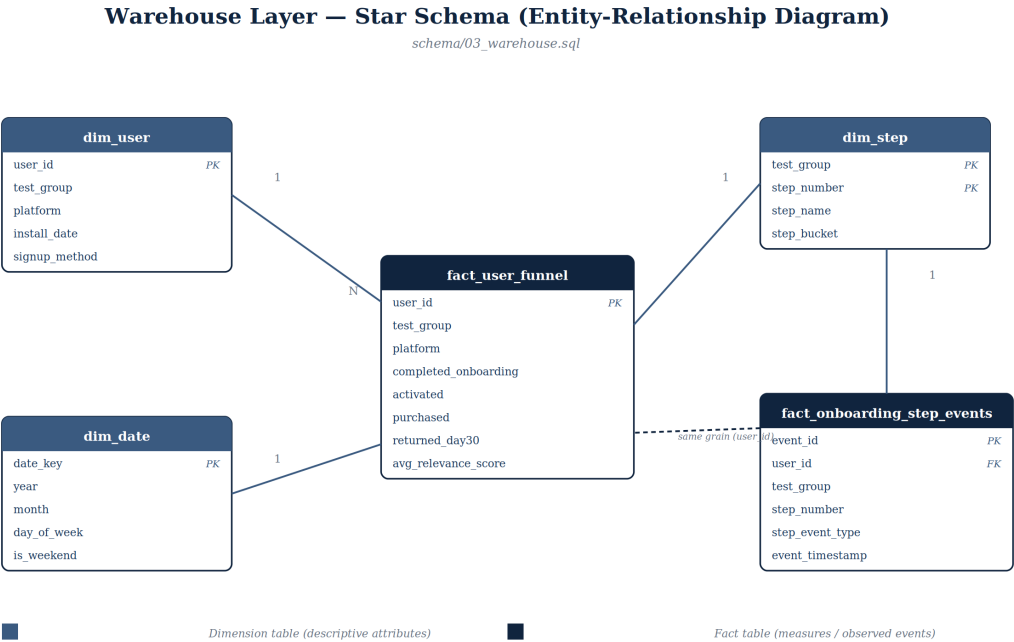
Staging performs three distinct transformations, applied in this order to avoid wasted work (validation and deduplication happen before the more expensive JSON-flattening step):

1. **Validation.** Rows failing any rule in Assignment 1’s Section 6 (`event_id / user_id / event_name / event_timestamp` non-null; `test_group` in {A,B,C}; `platform` in {iOS,Android}) are copied, with a `reject_reason`, into `stg_events_rejects` — a quarantine table, not a silent drop. This is a deliberate choice: silently dropping bad rows makes data-quality regressions invisible; a populated `stg_events_rejects` table is itself a monitorable signal (Section 5).
2. **Deduplication.** Rows are windowed by `event_id` (`ROW_NUMBER() OVER (PARTITION BY event_id ORDER BY _ingested_at)`) and only the first-ingested copy of each `event_id` is kept. This makes staging idempotent under at-least-once delivery, a standard assumption for client-side event collection (a network retry after an unacknowledged write is expected to produce a duplicate `event_id`, not a new one).
3. **Typed flattening.** `event_timestamp` is cast from text to a real SQLite datetime, and the fifteen most commonly queried `properties` keys are extracted into typed columns via `json_extract(...)`, so that no downstream SQL (warehouse or mart layer) needs to parse JSON. Properties not flattened here remain accessible in the retained raw `properties` column for ad hoc analysis.

3.3 Warehouse Layer (`schema/03_warehouse.sql`)

Table	Grain	Purpose
<code>dim_user</code>	1 row per <code>user_id</code>	Arm, platform, install date, account-creation status
<code>dim_date</code>	1 row per calendar date in the observed range	Generated via a recursive CTE; supports time-series joins for <code>mart_daily_funnel</code>
<code>dim_step</code>	1 row per (arm, step position)	Loaded directly from <code>stepmap.json</code> rather than derived from events, since it is experiment-design metadata, not observed behavior
<code>fact_onboarding_step_events</code>	1 row per step-level event (viewed/completed/abandoned)	Granular fact for step-level drop-off analysis
<code>fact_user_funnel</code>	1 row per <code>user_id</code>	The primary analytical fact; every KPI and guardrail in Assignment 1 is a <code>GROUP BY test_group</code> aggregation over this single table

`fact_user_funnel` is constructed with one `LEFT JOIN` from `dim_user` to `stg_events`, pivoting the long event stream into wide, typed boolean and timestamp columns via `MAX(CASE WHEN event_name = '...' THEN 1 ELSE 0 END)` idioms. This pivot is the single point in the pipeline where “did this user do X” becomes a queryable column rather than a row that must be searched for; every mart table and every future dashboard query builds on this fact rather than re-deriving it.



Warehouse star schema entity-relationship diagram

Figure 2. The warehouse’s dimensional model. `fact_user_funnel` is the primary fact (grain: one row per user); `fact_onboarding_step_events` is a secondary, finer-grained fact sharing the same `user_id`/`test_group` keys, retained separately because step-level drop-off analysis genuinely needs event-level granularity that the summary fact does not carry.

Indexing strategy. `dim_user.user_id` carries a unique index (it is the fact table’s join key); `fact_user_funnel.test_group` carries a non-unique index, since every mart query filters or groups by it. No index is placed on low-selectivity boolean flag columns, where a full scan is cheaper than an index lookup at this data volume.

3.4 Data Mart Layer (`schema/04_datamart.sql`)

Materialized (not virtual/view) tables, each built as a `GROUP BY` over `fact_user_funnel` or `fact_onboarding_step_events` , at the exact grain a dashboard or statistical test needs:

Mart Table	Grain	Consumed By
<code>mart_kpi_by_group</code>	1 row per arm	Primary/secondary KPI reporting; also the SRM check (Section 3, Assignment 1 event-tracking doc)
<code>mart_guardrails_by_group</code>	1 row per arm	Guardrail reporting
<code>mart_step_dropoff</code>	1 row per (arm, step)	Funnel/drop-off visualization
<code>mart_daily_funnel</code>	1 row per (date, arm)	Trend visualization; also supports a pre/post novelty-effect check

Marts are built as physical tables rather than SQL views specifically so that Assignment 4's dashboard queries are $O(\text{rows in mart})$, not $O(\text{rows in fact table})$ — a deliberate pre-aggregation trade-off, appropriate because the mart tables are small (rows = number of arms, or arms x steps) and rebuilding them is cheap relative to the cost of repeatedly aggregating the full fact table at dashboard-refresh time.

4. Ingestion Design

`build_db.py` orchestrates the pipeline as five ordered stages (schema creation, raw ingestion, staging, warehouse, mart), each executed via `sqlite3.Connection.executescript()` against the corresponding `.sql` file, with the Python layer responsible only for CSV parsing and JSON-file loading (`dim_step`) — the actual transformation logic lives in SQL, not in Python, so it is auditable independent of the orchestration code and portable to any SQL engine with `json_extract` support (e.g., PostgreSQL with minor syntax changes).

Ingestion is **idempotent by design**: re-running `build_db.py` against the same `events.csv` drops and rebuilds every table from scratch (the script deletes any pre-existing database file before connecting), so there is no notion of a partially-applied or double-applied ingestion run. This is an appropriate design for a batch, full-refresh pipeline at this data volume; Assignment 3 (pipeline automation) is where an incremental/append-only ingestion contract — necessary once the raw dataset no longer fits comfortably in a single full rebuild — would be introduced.

5. Data Quality Framework

Data quality is treated as a first-class, executable artifact (`schema/05_data_quality_checks.sql`), not an informal assertion in documentation. Twelve checks are run after every build and reported by `build_db.py`; they are organized here by the dimension of data quality each addresses:

Dimension	Checks
Completeness	<code>raw_row_count</code> , <code>users_missing_install_event</code>
Validity	<code>staging_rejects</code> (non-zero is expected and desired — Section 5.1), <code>invalid_test_group_values</code> , <code>kpi_rates_out_of_bounds</code>
Uniqueness	<code>staging_duplicate_event_ids</code>
Referential integrity	<code>orphan_fact_users</code> (every <code>fact_user_funnel</code> row must resolve to a <code>dim_user</code> row)
Logical / funnel-order consistency	<code>completed_without_started</code> , <code>activated_without_completed</code> , <code>purchased_without_activated</code> (funnel-stage ordering must be internally consistent — a user cannot be recorded as activated without first being recorded as having completed onboarding)
Structural	<code>mart_kpi_group_count</code> (must equal 3, the number of experiment arms), <code>negative_time_to_first_lesson</code>

5.1 Verifying the Validation Logic Itself

A pipeline whose validation checks always report zero problems is ambiguous: either the data is genuinely clean, or the validation logic is not actually being exercised. To distinguish these, `build_db.py` (by default) injects a small number of deliberately malformed rows directly into `raw_events` before staging runs — a null `event_id`, a null `user_id`, a null `event_name`, a null `event_timestamp`, an out-of-domain `test_group` value, an out-of-domain `platform` value, and a duplicated `event_id` — simulating the kind of partial/corrupted delivery a real client SDK can produce. A healthy run is expected to show `staging_rejects = 6` and a `raw_row_count` exceeding `stg_events` count by 9 (6 rejected plus 3 deduplicated), which is exactly what an observed run against the generated dataset shows (Section 6). This converts “the validation logic works” from an assumption into a repeatable, observed fact.

6. Observed Results

From a run against the 6,000-user, ~129,000-event generated dataset (Assignment 1), with dirty-row injection enabled:

Check	Result
<code>raw_row_count</code>	128,983
<code>staging_rejects</code>	6
<code>staging_duplicate_event_ids</code>	0
<code>users_missing_install_event</code>	0
<code>orphan_fact_users</code>	0
<code>completed_without_started</code>	0
<code>activated_without_completed</code>	0
<code>purchased_without_activated</code>	0
<code>invalid_test_group_values</code>	0
<code>mart_kpi_group_count</code>	3
<code>negative_time_to_first_lesson</code>	0
<code>kpi_rates_out_of_bounds</code>	0

Arm	Users Started	Onboarding Completion Rate	Activation Rate	Paid Conversion Rate	Day-30 Return (guardrail)
A (13 steps)	2,000	0.574	0.698	0.061	0.362
B (7 steps)	2,000	0.730	0.632	0.057	0.343
C (5 steps)	2,000	0.815	0.656	0.063	0.290

All twelve checks pass at their expected value (zero, except the two intentionally non-zero counts discussed in Section 5.1), and the resulting KPI table is directionally consistent with the hypothesis in the business-

understanding document: completion rate rises monotonically as step count falls, paid conversion remains approximately flat (~6%) across arms, and the Day-30 guardrail shows the expected trade-off for the shortest arm.

7. Assumptions and Limitations

1. **SQLite, not a production warehouse engine.** SQLite was chosen for zero-setup portability appropriate to a coursework deliverable. The SQL in `schema/` avoids SQLite-specific extensions other than `json_extract` (part of the SQL/JSON standard, also supported by PostgreSQL, MySQL 8+, and Snowflake), so migration to a production engine would primarily require changing the Python `sqlite3` connection layer, not rewriting the transformation SQL.
2. **Full-refresh, not incremental, ingestion.** Every run rebuilds every table from the full event history. This is appropriate at the current data volume (under 130,000 rows) but would not scale to a continuously arriving production stream; Assignment 3 is scoped to address incremental/scheduled ingestion.
3. **No slowly-changing-dimension (SCD) handling.** `dim_user` attributes (arm, platform) are treated as immutable for the lifetime of a user in this dataset, consistent with the “assigned once, never reassigned” randomization contract in the event-tracking plan. A production system with, e.g., platform migration (a user switching from Android to iOS) would need explicit SCD Type 2 handling, which is out of scope here.
4. **Single-database, single-file deployment.** No partitioning, replication, or concurrent-write handling is implemented, consistent with a batch, analyst-triggered build rather than a continuously operating service.

8. How to Run

```
python build_db.py \
  --events-csv ../assignment1/generated_data/events.csv \
  --stepmap-json ../assignment1/stepmap.json \
  --db db/ab_test.db
```

Produces `db/ab_test.db` (git-ignored; regenerate locally rather than committing a large binary artifact) and prints the data-quality report and KPI/guardrail preview shown in Section 6.

See the [assignment root README](#) for the rubric-to-deliverable mapping, and the [repository root README](#) for how this assignment fits into the full four-assignment pipeline.